

Week 1 – Day 3

Exponents and Logarithms

Objective

Review exponents and logarithms and understand why algorithms that repeatedly divide their input by two usually have logarithmic time complexity.

Exponent Rules

For a nonzero base a , the main exponent rules are:

1. Product rule:

$$a^m \cdot a^n = a^{m+n}$$

2. Quotient rule:

$$\frac{a^m}{a^n} = a^{m-n}$$

3. Power rule:

$$(a^m)^n = a^{mn}$$

4. Zero exponent:

$$a^0 = 1$$

5. Negative exponent:

$$a^{-n} = \frac{1}{a^n}$$

What Does a Logarithm Represent?

A logarithm answers the following question:

To what exponent must the base be raised to obtain a given number?

In general,

$$\log_b(x) = y \iff b^y = x$$

For example:

- $\log_2(8) = 3$ because $2^3 = 8$.
- $\log_2(32) = 5$ because $2^5 = 32$.
- $\log_2(1024) = 10$ because $2^{10} = 1024$

Connection with Binary Search

Starting with 32 elements and repeatedly dividing the search space by two gives:

$$32 \longrightarrow 16 \longrightarrow 8 \longrightarrow 4 \longrightarrow 2 \longrightarrow 1$$

It takes five divisions to reach one element, and:

$$\log_2(32) = 5$$

For 1024 elements, it takes ten divisions because:

$$\log_2(1024) = 10$$

Binary Search discards approximately half of the remaining elements during every iterations. Therefore, the number of iterations grows logarithmically as the input grows.

Time complexity: $O(\log n)$